

Clean-11 LOGO-CV Optimisation Results

SEP and MTOjects toy-object calibration, diagnostic deployment to 182 S4G galaxies, and production recommendation

Item	Value
Calibration sample	11 visually selected lower-contamination galaxies
Validation design	Leave-one-galaxy-out cross-validation with independent toy realisation
Deployment diagnostics	182 SEP + 182 MTOjects + 182 matched composite PNGs
Run identifier	clean11 logo_20260826_122730
Prepared	26 August 2026

Headline result

SEP produced a viable all-11 refit. The MTOjects all-11 refit produced no feasible toy recovery; therefore, the 182-galaxy MTOjects diagnostic deployment uses the feasible LOGO-CV winner rather than the failed refit.

1. Executive summary

The clean-11 experiment completed all 11 leave-one-galaxy-out folds for SEP and MTOjects. Each fold trained on 10 galaxies and assessed transfer to the excluded galaxy. Fold-derived candidates were also compared on a second deterministic injection realisation across all 11 galaxies. The selected parameters were then tested through standard eight-panel diagnostics across the complete 182-galaxy S4G sample.

Outcome	SEP	MTOjects
CV/final parameter source	All-11 refit	Feasible LOGO-CV winner (fold 9)
Toy detection rate on recorded selection set	31.82%	62.12%
Mean toy recall	32.10%	61.21%
Mean masked fraction on selection set	8.00%	12.16%
182-galaxy PNG batch	182/182 successful	182/182 successful

Interpretation

The optimisation metrics measure recovery of injected toys on the cleaner calibration sample. The 182-galaxy batch measures deployment behaviour and mask burden; it is not an independent truth-labelled accuracy test.

2. Parameter sets used

2.1 SEP all-11 refit

Parameter	Value	Parameter	Value
detect_thresh	1.0551	minarea	5
deblend_nthresh	64	deblend_cont	0.001201
back_size	32	filter_size	1
dilation_radius	1	max_area	7684
max_elongation	10.030	detect_on	original

2.2 MTOjects feasible LOGO-CV winner

Parameter	Value	Parameter	Value
move_factor	0.8632	min_distance	0.7217
gaussian_fwhm	0.4564	bg_variance	0.001413
minarea	4	dilation_radius	5
max_area	1755	max_elongation	3.754
winning_fold	9	detect_on	original

3. Optimisation results

Metric	SEP all-11 refit	MTOBJECTS LOGO-CV winner
Mean recall	30.34%	51.58%
Mean precision	0.21%	0.24%
Mean F-score	0.41%	0.49%
Mean toy recall	32.10%	61.21%
Toy detection rate	31.82%	62.12%
Mean masked fraction	8.00%	12.16%
Maximum masked fraction	11.28%	14.53%

3.1 MTOBJECTS all-11 refit failure

The separate 40-trial MTOBJECTS refit on all 11 calibration galaxies did not find a feasible candidate. Its nominal best trial had a toy detection rate of 0.00%, mean toy recall of 0.00%, and objective 59. It is retained for provenance but was not used for the diagnostic deployment.

Production decision

Use the feasible cross-validated MTOBJECTS winner for current science diagnostics. Do not promote the infeasible all-11 refit merely because it is labelled “best”; it is only the least-penalised member of a failed search.

4. Deployment across all 182 galaxies

Statistic	SEP	MTOBJECTS
Successful reports	182	182
Mean masked fraction	10.12%	11.47%
Median masked fraction	9.75%	11.47%
Minimum	5.12%	6.12%
Maximum	34.19%	26.72%
Galaxies above 15%	15	22
Galaxies above 20%	2	2

MTOBJECTS masked a larger fraction than SEP in 147 of 182 galaxies; SEP was larger in 35. The paired MTOBJECTS-minus-SEP difference averaged 1.34 percentage points, with a median of 1.23 percentage points.

5. Cleaner-sample deployment behaviour

Statistic	SEP	MTOObjects
Cleaner galaxies	11	11
Mean masked fraction	11.11%	12.16%
Median masked fraction	11.05%	12.40%
Range	8.00% to 15.80%	9.15% to 14.52%
Above 15%	1	0

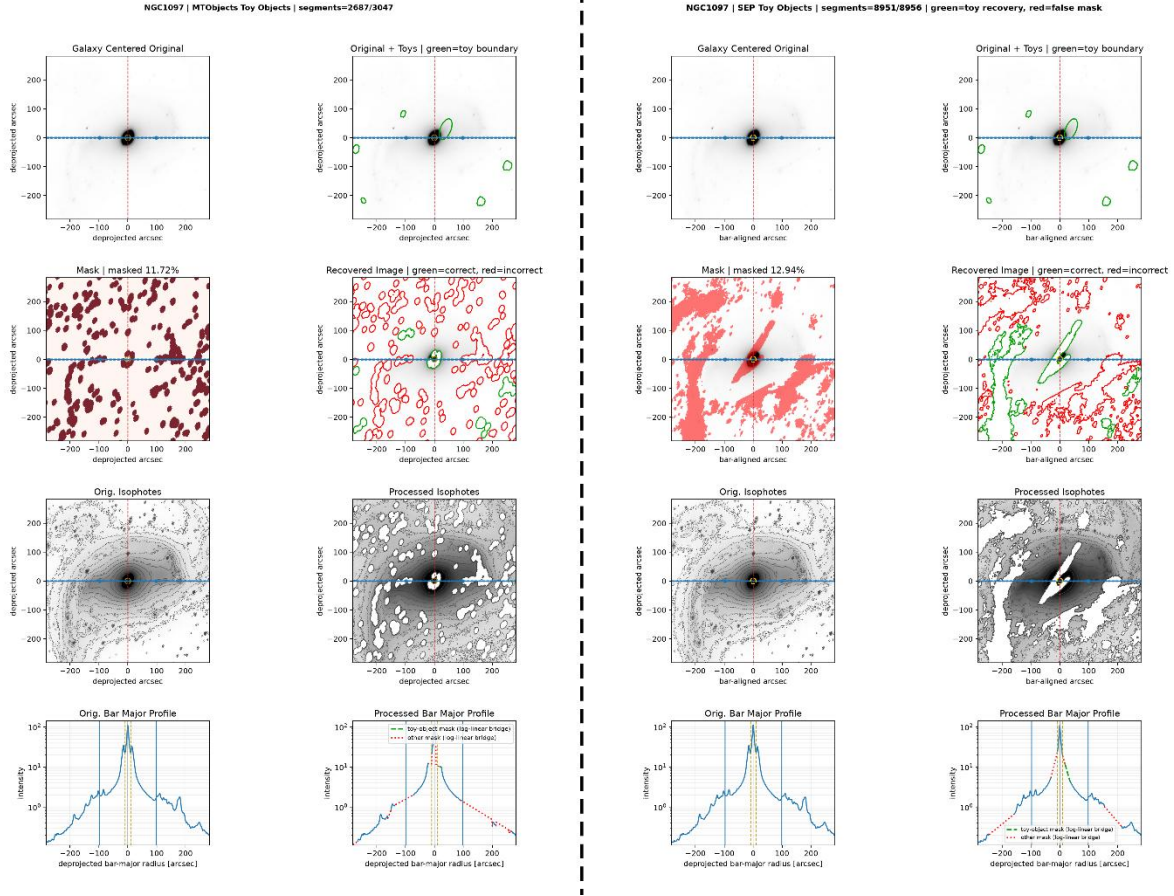


Figure 1. NGC1097 cleaner-sample composite. MTOObjects is shown on the left and SEP on the right; green boundaries identify injected toy locations/recovery annotations and red boundaries identify other masking.

6. Deployment outlier example

NGC1313 produced the largest masked fraction in both deployments: 34.19% for SEP and 26.72% for MTOObjects. This does not by itself prove incorrect masking, because the 182-galaxy set has no complete foreground-object truth labels. It does identify NGC1313 as a priority for visual review and sensitivity analysis.

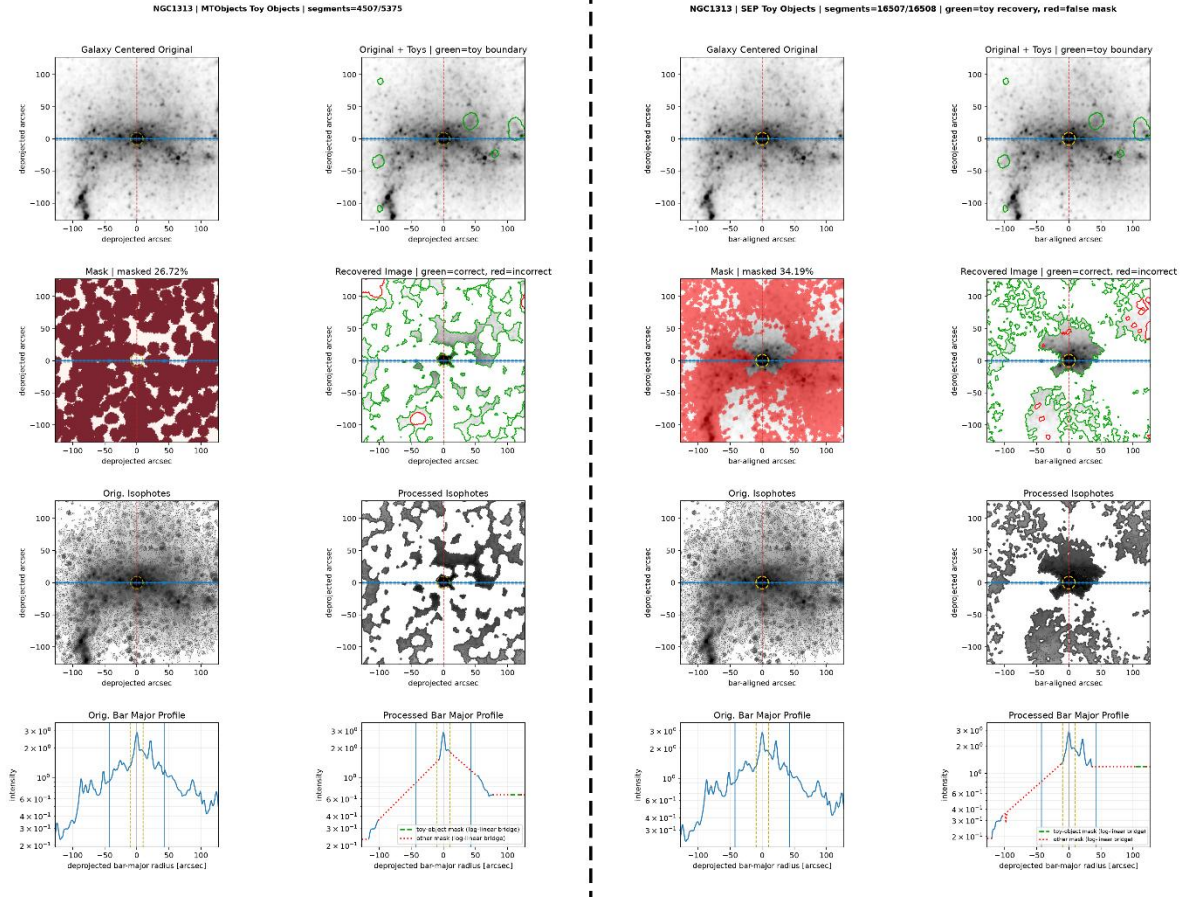


Figure 2. NGC1313 composite, the maximum-mask deployment case for both algorithms. The difference between the methods should be reviewed against real galaxy structure before adopting either mask uncritically.

7. Conclusions and recommendations

SEP: Retain the all-11 refit as the current clean-11 production candidate; it is feasible and completed the full diagnostic deployment.

MTOjects: Retain the feasible LOGO-CV winner as the current diagnostic candidate; reject the infeasible all-11 refit for production use.

Validation scope: Treat the 182 PNG sets as visual and mask-burden validation, not as a labelled estimate of foreground-removal accuracy.

Priority review: Review the 15 SEP and 22 MTOjects cases above 15% masked fraction, prioritising NGC1313 and the two cases above 20% for each method.

Science acceptance: Before freezing parameters for final measurements, quantify changes to bar profiles or other target measurements on high-mask and cleaner-sample subsets.

8. Output inventory

Product	Count	Location
SEP diagnostic PNGs	182	SEP\Toy Objects\clean11_logo_20260826_122730\PNG batch
MTOjects diagnostic PNGs	182	MTOjects\Toy Objects\clean11_logo_20260826_122730\PNG batch
Side-by-side composites	182	Toy Objects comparison\clean11_logo_20260826_122730
Cleaner-labelled PNGs	11 per method	Suffix: _clean

9. Reproducibility record

Both diagnostic batches used the original science image, six toys per galaxy, toy seed 202608299, peak amplitudes spanning 6–30 robust sigma, and one-pixel truth dilation. The calibration design used 11 galaxy-level folds and an independently seeded winner-selection injection set. All output counts and summary CSV statuses were checked after completion.